

Product Tensorization Resolves the Dimension Conjecture for the Random-Coordinate Rademacher LMC Scheme

Candidate solution packet for arXiv:1506.06966

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Status: candidate full solution, likely valid, pending human review. The theorem uses the strong-convexity sign that is required by the Gaussian example and by the proof of Proposition 12 in the source. The opposite sign printed in the proposition is a typographical error.

Abstract

Bonis conjectured that a random-coordinate Langevin scheme with Rademacher increments reaches Wasserstein error ϵ in $O(\epsilon^{-2}d^2 \log(d/\epsilon))$ scalar-coordinate operations under the hypotheses of Proposition 12 of [1]. We prove the intended conjecture. The key observation is that the all-pairs coordinate Lipschitz assumption in that proposition forces the potential to be a sum of one-dimensional potentials. The source’s proved one-dimensional $O(\sqrt{h})$ stationary-bias estimate therefore tensorizes to $O(\sqrt{dh})$. Combining this with the source’s random-coordinate contraction and taking $h \asymp \epsilon^2/d$ gives the asserted complexity.

1 Source problem and evidence

The source is Thomas Bonis, *Rates in the Central Limit Theorem and diffusion approximation via Stein’s Method*, arXiv:1506.06966v6 [1]. The scheme, Proposition 12, and the conjecture appear on PDF pages 16–17. At every step a uniform coordinate $I_n \in \{1, \dots, d\}$ and an independent Rademacher sign B_n are sampled, and

$$M_{n+1} = M_n + \left(-h \partial_{I_n} u(M_n) + \sqrt{2h} B_n \right) e_{I_n}. \quad (1)$$

The source conjectures that $h \asymp \epsilon^2/d$ and $n = O(\epsilon^{-2}d^2 \log(d/\epsilon))$ suffice.

The complete source evidence spans two pages:

of each step of the Euler-Maruyama discretization is of order d , the overall complexity of the algorithm is bounded by $O(\epsilon^{-2}d^2 \log(d/\epsilon))$. This rate can be improved whenever $\nabla^2 u$ is Lipschitz continuous, in which case one only requires $O(\epsilon^{-1}\sqrt{d} \log(d/\epsilon))$ steps to reach an accuracy ϵ , meaning the complexity of the algorithm is bounded by $O(\epsilon^{-1}d^{3/2} \log(d/\epsilon))$. One may wonder whether other discretization schemes could outperform the Euler-Maruyama scheme. However, the approach used to obtain the previous bounds are specific to the Euler-Maruyama scheme and cannot be used to evaluate the performance of another discretization scheme. Let us see how this can be done using our results. For instance, let $e_1, \dots, e_d \in \mathbb{R}^d$ be the canonical basis of $\mathbb{R}^d$, $(I_n)_{n \in \mathbb{N}}$ be independent uniform random variables on $\{1, \dots, d\}$, $(B_n)_{n \in \mathbb{N}}$ be independent Rademacher random variables and let us consider the discretization scheme $(M_n)_{n \in \mathbb{N}}$ with transitions

$$M_{n+1} = M_n + \left(-h \frac{\partial u}{\partial x_{I_n}}(M_n) + \sqrt{2h} B_n \right) e_{I_n}. \quad (10)$$

Following the computations presented in Section 10, we obtain the following result.

Proposition 12. *Let μ be a measure of $\mathbb{R}^d$ with density $f \in C^\infty(\mathbb{R}^d, \mathbb{R})$ and let $u = -\log f$. Suppose $\nabla u(0) = 0$ and assume there exists $\rho > 0, L > 0$ such that, for all $i \in \{1, \dots, d\}$ and $x, y \in \mathbb{R}^d$,*

$$\left(\frac{\partial u}{\partial x_i}(y) - \frac{\partial u}{\partial x_i}(x) \right) (y_i - x_i) \leq -\rho(y_i - x_i)^2 \quad (11)$$

and

$$\left| \frac{\partial u}{\partial x_i}(y) - \frac{\partial u}{\partial x_i}(x) \right| \leq L |y_i - x_i|. \quad (12)$$

Let $h > 0$ and let $(M_n)_{n \geq 0}$ be the Markov chain with $M_0 = 0$ and increments given by Equation (10). Then, there exist constants $C_1, C_2 > 0$ depending on ρ and L such that, for any $\epsilon > 0$, if $h = C_1 \epsilon^2 d^{-3}$ and $n = C_2 h^{-1} d \log(d/\epsilon)$, then the measure ν^n of M_n satisfies

$$W_2(\nu^n, \mu) \leq \epsilon.$$

Moreover, if μ is the Gaussian measure then the previous result holds true with $h = C_1 \epsilon^2 d^{-1}$.

Since each step of this one-dimensional discretization has a complexity independent of the dimension, the overall complexity of the LMC algorithm with the discretization scheme given by Equation (10) is bounded by $O(\epsilon^{-2}d^4 \log(d/\epsilon))$ and by $O(\epsilon^{-2}d^2 \log(d/\epsilon))$ when μ is the Gaussian measure. The discrepancy between the Gaussian case and the general case is due to the dependency on the dimension of the function f_k defined in Proposition 16 which we believe to be suboptimal, see Remark 17. Hence we conjecture the correct complexity of the LMC algorithm using the discretization scheme given by Equation (10) to be bounded by $O(\epsilon^{-2}d^2 \log(d/\epsilon))$ for target measures μ satisfying the assumptions

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of Proposition 12. Under this conjecture, this one-dimensional discretization scheme is still outperformed by the Euler-Maruyama scheme which reaches the complexity $O(\epsilon^{-1}d^{3/2} \log(d/\epsilon))$ whenever $\nabla^2 u$ is Lipschitz continuous. As this slightly stronger assumption should be verified in most practical cases, the Euler-Maruyama scheme should be more efficient than our one-dimensional scheme in practice.

5 Bounds for the Wasserstein distance of order 2: proofs of Theorems 2 and 9

Necessary sign correction

Write the target density as $f = Z^{-1}e^{-u}$. Equation (11) in Proposition 12 prints

$$(\partial_i u(y) - \partial_i u(x))(y_i - x_i) \leq -\rho(y_i - x_i)^2.$$

This cannot hold for the Gaussian target, even though the same proposition explicitly treats the Gaussian as an example. Together with the other hypotheses it is also incompatible with integrability of e^{-u} . Most decisively, the proof of Proposition 12 repeatedly uses the reverse strong-convexity inequality

$$(\partial_i u(y) - \partial_i u(x))(y_i - x_i) \geq \rho(y_i - x_i)^2. \quad (2)$$

We therefore formulate and solve the intended, nonvacuous conjecture using (2). This interpretation is a source correction, not an additional strengthening.

2 Definitions and source input

For probability measures with finite second moment on Euclidean space,

$$W_2(\alpha, \beta)^2 = \inf_{(X, Y)} \mathbb{E} \|X - Y\|^2,$$

where the infimum is over couplings with laws α and β .

We use one estimate already proved in [1]. Let $v : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $v(0) = 0$ and, for common constants $\rho, L > 0$,

$$(v(s) - v(t))(s - t) \geq \rho(s - t)^2, \quad |v(s) - v(t)| \leq L|s - t|. \quad (3)$$

Let μ_v have density proportional to $\exp(-\int_0^x v(r) dr)$, and let $\pi_{v,h}$ be the invariant law of

$$X^+ = X - hv(X) + \sqrt{2h} B, \quad \Pr(B = 1) = \Pr(B = -1) = \frac{1}{2}. \quad (4)$$

The proof of Proposition 12 in [1] obtains $W_2(\pi_h, \mu) \leq Cd^{3/2}\sqrt{h}$ before its special Gaussian refinement. Specializing that proved estimate to $d = 1$ gives constants $C_b, h_0 > 0$, depending only on ρ, L , such that

$$W_2(\pi_{v,h}, \mu_v) \leq C_b \sqrt{h} \quad (0 < h \leq h_0). \quad (5)$$

3 Proof intuition

The source's coordinate regularity assumption is not merely a coordinatewise smoothness bound along coordinate lines. It is asserted for every pair $x, y \in \mathbb{R}^d$:

$$|\partial_i u(y) - \partial_i u(x)| \leq L|y_i - x_i|.$$

When $x_i = y_i$, its right side vanishes. Thus $\partial_i u$ cannot depend on any coordinate other than x_i , and the target is a product measure. The source's one-dimensional bias estimate (5) can then be applied separately to every factor. Quadratic transport cost is additive on products, so the squared biases add and produce $C_b \sqrt{dh}$ rather than the source's general $Cd^{3/2}\sqrt{h}$ bound. The already available random-scan contraction then supplies the burn-in estimate.

4 The theorem

Theorem 1 (Resolution of the intended conjecture). *Let $d \geq 1$, let $u \in C^\infty(\mathbb{R}^d)$ satisfy $\nabla u(0) = 0$, and suppose $Z^{-1}e^{-u}$ is a probability density. Assume that there are $\rho, L > 0$ such that, for every $i \in \{1, \dots, d\}$ and every $x, y \in \mathbb{R}^d$,*

$$(\partial_i u(y) - \partial_i u(x))(y_i - x_i) \geq \rho(y_i - x_i)^2, \quad (6)$$

$$|\partial_i u(y) - \partial_i u(x)| \leq L|y_i - x_i|. \quad (7)$$

Let (M_n) be the chain (1) with $M_0 = 0$, and write $\nu_n = \text{Law}(M_n)$ and $\mu(dx) = Z^{-1}e^{-u(x)}dx$.

There are constants $a, A > 0$, depending only on ρ, L , such that for every $0 < \epsilon \leq 1$, the choice

$$h = a\frac{\epsilon^2}{d}, \quad n \geq A\epsilon^{-2}d^2 \log\left(\frac{2d}{\epsilon}\right)$$

implies

$$W_2(\nu_n, \mu) \leq \epsilon.$$

Consequently the scalar-gradient arithmetic complexity is $O_{\rho,L}(\epsilon^{-2}d^2 \log(d/\epsilon))$.

Proof. Step 1: the assumptions force a product potential. Fix i . If $x_i = y_i$, then (7) gives $\partial_i u(x) = \partial_i u(y)$. Hence a one-dimensional function v_i exists such that

$$\partial_i u(x) = v_i(x_i) \quad (x \in \mathbb{R}^d).$$

Put $u_i(t) = \int_0^t v_i(s) ds$. The function $u(x) - \sum_i u_i(x_i)$ has zero gradient and is constant. Absorbing that constant into the normalizing factor,

$$\mu = \bigotimes_{i=1}^d \mu_i, \quad \mu_i(dt) = Z_i^{-1} e^{-u_i(t)} dt. \quad (8)$$

Conditions (6)–(7) become

$$(v_i(s) - v_i(t))(s - t) \geq \rho(s - t)^2, \quad |v_i(s) - v_i(t)| \leq L|s - t|. \quad (9)$$

Also $v_i(0) = 0$. Strong convexity ensures that every Z_i is finite.

Step 2: coordinate invariant laws and their bias. Let $K_{i,h}$ be the one-dimensional kernel

$$x \mapsto x - hv_i(x) + \sqrt{2h} B.$$

For a synchronous choice of B , (9) gives

$$|s - t - h(v_i(s) - v_i(t))|^2 \leq (1 - 2\rho h + L^2 h^2)|s - t|^2. \quad (10)$$

Thus, when $0 < h < 2\rho/L^2$, $K_{i,h}$ is a strict Wasserstein contraction and has a unique invariant law $\pi_{i,h}$. Applying the source estimate (5) to each v_i yields, uniformly in i ,

$$W_2(\pi_{i,h}, \mu_i) \leq C_b \sqrt{h}. \quad (11)$$

Set $\Pi_h = \bigotimes_i \pi_{i,h}$. Updating coordinate i preserves Π_h because it replaces its i th marginal by the same invariant marginal and leaves the other marginals untouched. Hence Π_h is invariant under the random-scan kernel of (1).

For product probability measures and squared Euclidean cost,

$$W_2\left(\bigotimes_i \alpha_i, \bigotimes_i \beta_i\right)^2 = \sum_i W_2(\alpha_i, \beta_i)^2. \quad (12)$$

Indeed, every joint coupling induces a coordinate coupling and hence has cost at least the sum on the right; the product of coordinatewise optimal (or arbitrarily near-optimal) couplings gives the reverse inequality. Combining (8), (11), and (12),

$$W_2(\Pi_h, \mu) \leq C_b \sqrt{dh}. \quad (13)$$

Step 3: random-scan contraction and initial moment. Couple two copies of one chain step using the same coordinate I and sign B . Summing (10) over the uniformly selected coordinate gives

$$\mathbb{E}[\|X^+ - Y^+\|^2 \mid X = x, Y = y] \leq \left(1 - \frac{2\rho h - L^2 h^2}{d}\right) \|x - y\|^2. \quad (14)$$

In particular, the invariant law is unique and equals Π_h . If $h \leq \rho/L^2$, then $2\rho h - L^2 h^2 \geq \rho h$, and iteration gives

$$W_2(\nu_n, \Pi_h) \leq e^{-\rho h n/(2d)} W_2(\delta_0, \Pi_h). \quad (15)$$

It remains to bound the last factor. If $X_i \sim \pi_{i,h}$, stationarity of (4), independence and centering of B , and $v_i(0) = 0$ give

$$\begin{aligned} 0 &= \mathbb{E}[(X_i - h v_i(X_i) + \sqrt{2h}B)^2 - X_i^2] \\ &= \mathbb{E}[-2hX_i v_i(X_i) + h^2 v_i(X_i)^2 + 2h]. \end{aligned}$$

Using (9),

$$(2\rho - L^2 h) \mathbb{E} X_i^2 \leq 2.$$

For $h \leq \rho/L^2$ this yields

$$W_2(\delta_0, \Pi_h)^2 = \sum_i \mathbb{E} X_i^2 \leq \frac{2d}{\rho}. \quad (16)$$

Step 4: choose the parameters. Let

$$a \leq \min \left\{ h_0, \frac{\rho}{L^2}, \frac{1}{16C_b^2}, 1 \right\}$$

be a fixed positive constant and set $h = a\epsilon^2/d$. Then (13) gives

$$W_2(\Pi_h, \mu) \leq \epsilon/4.$$

By (15)–(16), it is enough to take

$$n \geq \frac{2d}{\rho h} \log \left(\frac{2\sqrt{2d/\rho}}{\epsilon} \right)$$

to obtain $W_2(\nu_n, \Pi_h) \leq \epsilon/2$. The triangle inequality then gives $W_2(\nu_n, \mu) \leq 3\epsilon/4 < \epsilon$. Since $h = a\epsilon^2/d$, the displayed lower bound for n is $O_{\rho,L}(\epsilon^{-2}d^2 \log(d/\epsilon))$. $\square$

5 Verification notes and limitations

The proof after (5) is elementary. The most important checks are the following.

1. The separation argument uses the fact that the source states its coordinate Lipschitz condition for all $x, y \in \mathbb{R}^d$, not merely for pairs that differ only in coordinate i .
2. The constant in the source estimate (5) depends only on the common parameters ρ, L when $d = 1$, so it is uniform over i .
3. The product W_2 identity (12) is exact because the squared Euclidean cost is the sum of coordinate costs.
4. No independence of the coordinate update counts is assumed. Mixing is handled directly by the random-scan contraction (14).
5. The theorem answers the intended corrected-sign conjecture. With the negative sign literally printed in the source, the class is inconsistent with the Gaussian clause and the intended statement is vacuous.

No computational experiment is used. The packet also contains an adversarial verifier report.

6 Bounded novelty check

On 9 August 2026, bounded searches used the exact source title and arXiv id, the quoted complexity, and combinations of “random coordinate Langevin”, “Rademacher”, “product”, and “tensorization”. Later work of Ding–Li–Lu–Wright [2] studies a related random-coordinate LMC with Gaussian coordinate noise and more general potentials. The search did not locate an explicit resolution of this exact Rademacher-noise conjecture or the product implication of Bonis’s all-pairs coordinate assumption. This is not proof of novelty; novelty confidence is moderate.

7 Human review recommendation

The packet should be reviewed by a probability or MCMC specialist. Priority checks are the source sign correction, uniformity of the dimension-one bias constant, and whether the same product observation occurs in later exact-scheme literature. Subject to those checks, the mathematical verdict is likely valid.

References

- [1] T. Bonis, *Rates in the Central Limit Theorem and diffusion approximation via Stein’s Method*, arXiv:1506.06966v6, 2018. <https://arxiv.org/abs/1506.06966>
- [2] Z. Ding, Q. Li, J. Lu, and S. J. Wright, *Random Coordinate Langevin Monte Carlo*, Proceedings of Machine Learning Research 134 (COLT 2021), 1683–1710; arXiv:2010.01405. <https://arxiv.org/abs/2010.01405>